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Department of Computer Science and Engineering (Artificial Intelligence and Machine Learning)

Date: 17/06/2026	Test – III	Max. Marks: 50 + 10
Semester: II	UG	Duration: 2 Hours
Course Title: AI Foundations for Engineers		Course Code: CI124TA

Part – A

Q. No.	Scheme and Solutions			M
1.	Task	Suitable Technique	Justification	2
	Detect faces in images	CNN	CNN extracts spatial features and patterns from images.	
	Translate speech into text	RNN	RNN processes sequential data and captures temporal dependencies.	
	Learn optimal delivery routes	Reinforcement Learning	RL learns optimal actions through rewards and penalties.	
	<i>Correct mapping of techniques to tasks – 1 Mark</i> <i>Appropriate justification – 1 Mark</i>			
2.	Input → Hidden: - Weights = $3 \times 2 = 6$ - Biases for hidden neurons = 2 Total = 8 Hidden → Output: - Weights = $2 \times 1 = 2$ - Bias for output neuron = 1 Total = 3 Total weights including biases = $8 + 3 = 11$ <i>Hidden layer weight calculation – 1 Mark</i> <i>Output layer weight calculation and final answer – 1 Mark</i>			2
3.	i. Possible source of bias: - Biased or unrepresentative training data. - Historical hiring data may underrepresent applicants from that region. ii. Lifecycle stage responsible: - Data collection and data preparation stage. iii. Impact on stakeholders: - Unfair rejection of qualified candidates. - Loss of trust in the AI system. - Legal and ethical concerns for the organization. <i>Source of bias identified – 1 Mark</i> <i>Lifecycle stage and stakeholder impact – 1 Mark</i>			2

4.	System B (RNN) is likely to perform better. Justification: • Stock prices form a time-series sequence. • RNN maintains information from previous time steps through its hidden state. • Feedforward networks treat inputs independently and cannot naturally model temporal dependencies. <i>Correct identification of suitable model – 1 Mark</i> <i>Justification based on sequence dependency – 1 Mark</i>	2
5.	i. Possible data issues: • Historical hiring data contains gender bias. • Imbalanced dataset with fewer successful female candidates. • Gender-correlated features influencing decisions. ii. Fairness concerns: • Gender discrimination. • Violation of equal opportunity and fairness principles. • Ethical and legal implications due to biased decision-making. <i>Data issue – 1 Mark</i> <i>Fairness issue – 1 Mark</i>	2

Part – B

Q. No.	Scheme and Solutions	M
1a.	Structure of an Artificial Neuron An artificial neuron consists of: • Inputs ($x_1, x_2, \dots, x_n$) • Weights ($w_1, w_2, \dots, w_n$) • Bias (b) • Summation unit • Activation function • Output (y) Mathematical Model Weighted sum: $z = \sum_{i=1}^n w_i x_i + b$ Output: $y = f(z)$ where: • x_i= input • w_i= weight • b= bias • $f(z)$= activation function Activation Function The activation function determines whether a neuron should be activated. Examples: • Sigmoid • Tanh • ReLU Effect on Decision Boundary • Without activation, the neuron can only model a linear decision boundary. • Non-linear activation functions enable complex and non-linear decision boundaries.	5

	This improves classification capability. <table><tr><td>Structure of artificial neuron</td><td>1</td></tr><tr><td>Mathematical model</td><td>2</td></tr><tr><td>Activation function</td><td>1</td></tr><tr><td>Effect on decision boundary</td><td>1</td></tr></table>	Structure of artificial neuron	1	Mathematical model	2	Activation function	1	Effect on decision boundary	1			
Structure of artificial neuron	1											
Mathematical model	2											
Activation function	1											
Effect on decision boundary	1											
1b.	Case 1: $b = 0.1$ $z = (1)(0.2) + (0)(-0.3) + (1)(0.4) + 0.1$ $z = 0.7$ Sigmoid: $\sigma(z) = \frac{1}{1 + e^{-z}}$ $\sigma(0.7) = 0.668$ Output = 0.668 Case 2: $b = 0.5$ $z = (1)(0.2) + (0)(-0.3) + (1)(0.4) + 0.5$ $z = 1.1$ $\sigma(1.1) = 0.750$ Output = 0.750 Comment Increasing the bias increases the net input and therefore increases the neuron output. <table><tr><td>Net input calculation</td><td>1</td></tr><tr><td>Sigmoid output</td><td>1</td></tr><tr><td>New net input</td><td>1</td></tr><tr><td>New output</td><td>1</td></tr><tr><td>Comment on bias effect</td><td>1</td></tr></table>	Net input calculation	1	Sigmoid output	1	New net input	1	New output	1	Comment on bias effect	1	5
Net input calculation	1											
Sigmoid output	1											
New net input	1											
New output	1											
Comment on bias effect	1											
2.	Step 1: Forward Pass Given: $x_1 = 0.35, x_2 = 0.9$ $w_{13} = 0.1, w_{23} = 0.8$ $w_{14} = 0.4, w_{24} = 0.6$ $w_{35} = 0.3, w_{45} = 0.9$ Hidden Neuron H3 $net_3 = (0.35)(0.1) + (0.9)(0.8)$ $net_3 = 0.755$ $y_3 = \sigma(0.755) = 0.680$ Hidden Neuron H4 $net_4 = (0.35)(0.4) + (0.9)(0.6)$ $net_4 = 0.680$ $y_4 = \sigma(0.680) = 0.664$ Output Neuron $net_5 = (0.680)(0.3) + (0.664)(0.9)$ $net_5 = 0.8016$ $y_5 = \sigma(0.8016) = 0.690$ Step 2: Output Error $\delta_5 = (T - y_5)y_5(1 - y_5)$ $= (0.5 - 0.690)(0.690)(0.310)$ $\delta_5 = -0.0406$ Step 3: Hidden Layer Error	10										

$$\delta_3 = y_3(1 - y_3)(w_{35}\delta_5)$$

$$\delta_3 = -0.00266$$

$$\delta_4 = y_4(1 - y_4)(w_{45}\delta_5)$$

$$\delta_4 = -0.00815$$

Step 4: Weight Updates

$$w_{35}^{new} = 0.3 + 0.1(-0.0406)(0.680)$$

$$w_{35}^{new} = 0.2972$$

$$w_{45}^{new} = 0.9 + 0.1(-0.0406)(0.664)$$

$$w_{45}^{new} = 0.8973$$

$$w_{13}^{new} = 0.0999$$

$$w_{23}^{new} = 0.7998$$

$$w_{14}^{new} = 0.3997$$

$$w_{24}^{new} = 0.5993$$

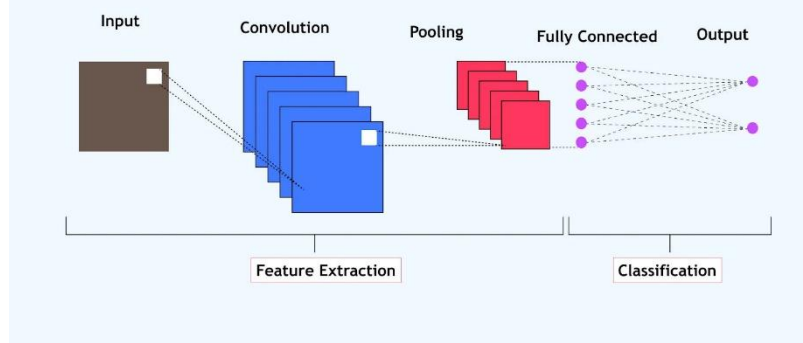
Updated Weights

$$w_{13} = 0.0999, w_{23} = 0.7998, w_{14} = 0.3997, w_{24} = 0.5993, w_{35} = 0.2972, w_{45} = 0.8973$$

Hidden layer forward pass	2
Output layer forward pass	2
Output layer error	2
Hidden layer error	2
Weight updates	2

3

The Architecture of Convolutional Neural Networks



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Input Layer

Receives image pixels as input.

Convolution Layer

Applies filters/kernels to extract features such as edges, corners, textures, and shapes.

Activation Function

Introduces non-linearity.

Commonly used:

- ReLU
- Sigmoid
- Tanh

Pooling Layer

Reduces spatial dimensions and computational cost.

Types:

- Max Pooling
- Average Pooling

Fully Connected Layer

Combines extracted features and performs classification.

	Output Layer Produces final prediction. • Sigmoid for binary classification • Softmax for multiclass classification Advantages • Automatic feature extraction • Parameter sharing • Translation invariance <table><tr><td><i>CNN architecture diagram</i></td><td><i>2</i></td></tr><tr><td><i>Input layer</i></td><td><i>1</i></td></tr><tr><td><i>Convolution layer</i></td><td><i>2</i></td></tr><tr><td><i>Activation layer</i></td><td><i>1</i></td></tr><tr><td><i>Pooling layer</i></td><td><i>2</i></td></tr><tr><td><i>Fully connected layer</i></td><td><i>1</i></td></tr><tr><td><i>Output layer</i></td><td><i>1</i></td></tr></table>	<i>CNN architecture diagram</i>	<i>2</i>	<i>Input layer</i>	<i>1</i>	<i>Convolution layer</i>	<i>2</i>	<i>Activation layer</i>	<i>1</i>	<i>Pooling layer</i>	<i>2</i>	<i>Fully connected layer</i>	<i>1</i>	<i>Output layer</i>	<i>1</i>	
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<i>Output layer</i>	<i>1</i>															
4a	i) Data Collection and Processing Data Collection and Processing are crucial stages in the Responsible AI Lifecycle because AI systems depend heavily on the quality of training data. Data Quality and Representativeness • Data should be accurate, complete, relevant, and representative of all stakeholder groups. • Representative datasets reduce the risk of unfair outcomes. Bias Identification and Mitigation • Historical and societal biases present in datasets must be identified. • Bias mitigation techniques should be applied before model training. Privacy, Security and Preprocessing • Data cleaning, normalization, and validation improve reliability. • Personal information should be protected through appropriate privacy measures and governance policies. (ii) Importance of Testing and Monitoring Testing Before Deployment • Verifies correctness, fairness, robustness, and reliability. • Identifies risks, biases, and performance issues before deployment. • Ensures compliance with ethical and regulatory requirements. Monitoring After Deployment • Detects model drift and data drift. • Tracks fairness, accuracy, and system performance. • Supports continuous improvement and retraining when required. <table><tr><td><i>Data quality and representativeness</i></td><td><i>1</i></td></tr><tr><td><i>Bias identification and mitigation</i></td><td><i>1</i></td></tr><tr><td><i>Privacy, security and preprocessing</i></td><td><i>1</i></td></tr><tr><td><i>Testing before deployment</i></td><td><i>1.5</i></td></tr><tr><td><i>Monitoring after deployment</i></td><td><i>1.5</i></td></tr></table>	<i>Data quality and representativeness</i>	<i>1</i>	<i>Bias identification and mitigation</i>	<i>1</i>	<i>Privacy, security and preprocessing</i>	<i>1</i>	<i>Testing before deployment</i>	<i>1.5</i>	<i>Monitoring after deployment</i>	<i>1.5</i>	6				
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<i>Monitoring after deployment</i>	<i>1.5</i>															
4b	1. Evasion Attack • Adversarial inputs are crafted during testing/inference. • Objective is to force incorrect predictions. 2. Poisoning Attack • Training data is deliberately manipulated. • Objective is to corrupt the learning process or degrade performance.	4														

3. White-Box Attack

- Attacker has complete knowledge of the model architecture, parameters, and training process.
- Enables highly effective attacks.

4. Black-Box Attack

- Attacker has access only to model inputs and outputs.
- Model behavior is inferred through repeated queries.

Classification

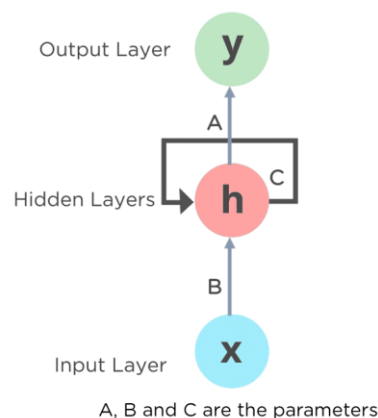
- Evasion and Poisoning attacks describe **attacker goals**.
- White-box and Black-box attacks describe **attacker capabilities and knowledge**.

<i>Evasion Attack</i>	<i>1</i>
<i>Poisoning Attack</i>	<i>1</i>
<i>White-box Attack</i>	<i>1</i>
<i>Black-box Attack</i>	<i>1</i>

5a

A Recurrent Neural Network (RNN) is a neural network designed for processing sequential data. Unlike a feedforward neural network, an RNN contains a **recurrent connection** through which information from the previous time step is passed to the current time step. The hidden state acts as memory and stores information about previous inputs in the sequence.

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At each time step:

1. The current input x_t is received.
2. The previous hidden state h_{t-1} is combined with the current input.
3. A new hidden state h_t is generated.
4. The output y_t is produced.

Mathematical Representation

Hidden State:

$$h_t = f(W_{xh}x_t + W_{hh}h_{t-1} + b_h)$$

Output:

$$y_t = g(W_{hy}h_t + b_y)$$

where:

- W_{xh} = Input-to-hidden weights
- W_{hh} = Recurrent weights
- W_{hy} = Hidden-to-output weights
- b_h, b_y = Bias terms

Working

The hidden state h_t carries information from previous time steps, enabling the network to retain contextual information while processing a sequence. Thus, the output at a given time step depends

	on both the current input and previous hidden states.		
	<i>Neat diagram of RNN architecture</i>	<i>1</i>	
	<i>Explanation of recurrent connection and hidden state</i>	<i>2</i>	
	<i>Mathematical representation</i>	<i>1</i>	
	<i>Working of RNN across time steps</i>	<i>1</i>	
5b	Reinforcement Learning Reinforcement Learning (RL) is a learning paradigm in which an agent learns an optimal policy by interacting with an environment and receiving rewards or penalties. Smart Irrigation Scenario Agent - Smart irrigation controller. Environment - Farm field, crops, weather conditions, and soil. State - Soil moisture level. - Temperature. - Humidity. - Rainfall forecast. Action - Increase water supply. - Decrease water supply. - Stop irrigation. Reward - Positive reward for healthy crop growth and efficient water usage. - Negative reward for overwatering or underwatering. Policy - A strategy that maps environmental states to irrigation actions to maximize long-term cumulative reward. Conclusion: The agent learns the optimal watering strategy through repeated interaction with the environment and feedback in the form of rewards.		5
	<i>Reinforcement Learning definition</i>	<i>1</i>	
	<i>Agent</i>	<i>1</i>	
	<i>Environment</i>	<i>1</i>	
	<i>State and Action</i>	<i>1</i>	
	<i>Reward and Policy</i>	<i>1</i>	